課題 2

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1 HDL

1.1 HalfAdder

```
CHIP HalfAdder {
    IN a, b;
    OUT sum,
        carry;

    PARTS:
    And(a=a,b=b,out=carry);
    Xor(a=a,b=b,out=sum);
}
```

1.2 FullAdder(HalfAdder*2)

```
CHIP FullAdder {
    IN a, b, c;
    OUT sum,
        carry;

    PARTS:
    HalfAdder(a=a,b=b,sum=sum1,carry=carry1);
    HalfAdder(a=sum1,b=c,sum=sum,carry=carry2);
    Or(a=carry1,b=carry2,out=carry);

}
```

1.3 FullAdder(without HalfAdder)

```
CHIP FullAdder {
    IN a, b, c;
    OUT sum,
        carry;

    PARTS:
```

```
    And(a=a,b=b,out=carry1);
    Xor(a=a,b=b,out=sum1);
    And(a=sum1,b=c,out=carry2);
    Xor(a=sum1,b=c,out=sum);
    Or(a=carry1,b=carry2,out=carry);
}

```

1.4 Add16

```
CHIP Add16 {
    IN a[16], b[16];
    OUT out[16];

    PARTS:
    HalfAdder(a=a[0],b=b[0],sum=out[0],carry=temp0);
    FullAdder(a=a[1],b=b[1],c=temp0,sum=out[1],carry=temp1);
    FullAdder(a=a[2],b=b[2],c=temp1,sum=out[2],carry=temp2);
    FullAdder(a=a[3],b=b[3],c=temp2,sum=out[3],carry=temp3);
    FullAdder(a=a[4],b=b[4],c=temp3,sum=out[4],carry=temp4);
    FullAdder(a=a[5],b=b[5],c=temp4,sum=out[5],carry=temp5);
    FullAdder(a=a[6],b=b[6],c=temp5,sum=out[6],carry=temp6);
    FullAdder(a=a[7],b=b[7],c=temp6,sum=out[7],carry=temp7);
    FullAdder(a=a[8],b=b[8],c=temp7,sum=out[8],carry=temp8);
    FullAdder(a=a[9],b=b[9],c=temp8,sum=out[9],carry=temp9);
    FullAdder(a=a[10],b=b[10],c=temp9,sum=out[10],carry=temp10);
    FullAdder(a=a[11],b=b[11],c=temp10,sum=out[11],carry=temp11);
    FullAdder(a=a[12],b=b[12],c=temp11,sum=out[12],carry=temp12);
    FullAdder(a=a[13],b=b[13],c=temp12,sum=out[13],carry=temp13);
    FullAdder(a=a[14],b=b[14],c=temp13,sum=out[14],carry=temp14);
    FullAdder(a=a[15],b=b[15],c=temp14,sum=out[15],carry=temp15);
}
```

1.5 Incrementer

```
CHIP Inc16 {
    IN in[16];
    OUT out[16];

    PARTS:
    Add16(a=in,b[0]=true,out=out);
}
```

1.6 Alu

```

CHIP ALU {

```

```
4
5       IN
6           x[16], y[16],
7           zx, nx, zy, ny, f, no;
8       OUT
9           out[16], zr, ng;
10
11      PARTS:
12
13      Mux16(a=x, b=false, sel=zx, out=x1);
14      Not16(in=x1, out=notx);
15      Mux16(a=x1, b=notx, sel=nx, out=x2);
16
17      Mux16(a=y, b=false, sel=zy, out=y1);
18      Not16(in=y1, out=noty);
19      Mux16(a=y1, b=noty, sel=ny, out=y2);
20
21      Add16(a=x2, b=y2, out=addOut);
22      And16(a=x2, b=y2, out=andOut);
23      Mux16(a=andOut, b=addOut, sel=f, out=resultOut);
24
25      Not16(in=resultOut, out=notresult);
26      Mux16(a=resultOut, b=notresult, sel=no, out=out1);
27
28      And16(a=out1,b=true,out=out);
29
30      Or16Way(in=out1,out=zr1);
31      Not(in=zr1,out=zr);
32
33      IsNeg(in=out1,out=ng);
34
35  }
```

1.7 helpchip

1.7.1 Or16Way

```
1   CHIP Or16Way {
2       IN in[16];
3       OUT out;
4
5       PARTS:
6       Or(a = in[0], b = in[1], out = o1);
7       Or(a = o1, b = in[2], out = o2);
8       Or(a = o2, b = in[3], out = o3);
9       Or(a = o3, b = in[4], out = o4);
10      Or(a = o4, b = in[5], out = o5);
11      Or(a = o5, b = in[6], out = o6);
12      Or(a = o6, b = in[7], out = o7);
13      Or(a = o7, b = in[8], out = o8);
```

```
    Or(a = o8, b = in[9], out = o9);
    Or(a = o9, b = in[10], out = o10);
    Or(a = o10, b = in[11], out = o11);
    Or(a = o11, b = in[12], out = o12);
    Or(a = o12, b = in[13], out = o13);
    Or(a = o13, b = in[14], out = o14);
    Or(a = o14, b = in[15], out = out);
}
```

1.7.2 IsNeg

```
CHIP IsNeg {

    IN in[16];
    OUT out;

    PARTS:
    Or(a = in[15], b = false, out = out);
}
```

2 加算器および Alu の実装についての説明

2.1 add16

まず、HalfAdder に入力をし、FullAdder を繰り返し呼び出すことで、add16 を実装できた。ただし、オーバーフローを無視する。

2.2 Alu

x,y の演算をそれぞれ実装して、制御ビットを実装する。また、bliut chip がない chip を定義し直す。